

1. Suppose you are building a stock tracking application that allows users to track the prices of different stocks. You decide to use a dictionary to store the stock information, where each stock symbol is the key and the corresponding value is a dictionary that contains the stock name, price, and other details.

Your application should allow users to add new stocks to track, remove existing stocks, and update the prices of stocks.

Write a function in Python that takes in a stock symbol and a new price and updates the price for that stock. The function should also print a message confirming the update.

2. Suppose you are working on a project that requires you to store information about various products in a dictionary. Each product is identified by a unique product ID, and the corresponding value in the dictionary is a dictionary that contains details about the product, such as its name, description, price, and stock level.

You need to write a function in Python that takes in the dictionary of products and returns the product ID of the product with the lowest price. If there are multiple products with the same lowest price, the function should return the product ID of the first such product encountered.

3. You work for a social media company and have been tasked with analyzing user engagement on the platform. You have been given a dataset of user interactions for the past month, which includes information such as user ID, post ID, timestamp, and type of interaction (e.g. like, comment, share).

Your goal is to create a report that summarizes the user engagement on the platform.

Specifically, you need to calculate the total number of interactions for each user and for each post, and also identify the most popular posts based on the number of interactions they received.

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